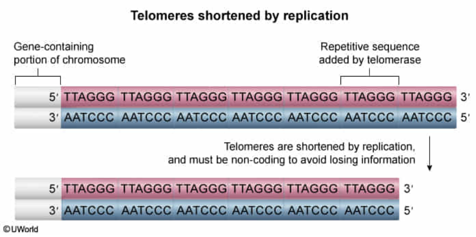
The DNA of telomeres can be most accurately described as:

- ✔ A. noncoding and highly repetitive. [89%]
- B. single-stranded and self-complementing. [3%]
- C. able to encode multiple unique proteins. [2%]
- D. easily cleaved by restriction enzymes. [4%]

Correct
90% answered correctly

Explanation:



Due to a phenomenon called the end-replication problem, the ends of a chromosome cannot be completely replicated, so a chromosome **shortens slightly** after each replication event. This shortening occurs at the ends of the chromosome, called the telomeres. Telomeres are **noncoding** DNA sequences that **protect the rest of the chromosome** from degradation by being slowly degraded themselves. If telomeres encoded proteins, some of the coding information would be lost each time the cell replicated, which would be harmful to the cell. For this reason, telomeres do not encode any proteins **(Choice C)**.

When telomerase extends telomeres, it does so by adding the sequence 5′-TTAGGG-3′ to the end of one of the strands on the chromosome multiple times in succession. DNA polymerase then adds the complementary bases to the other strand to generate double-stranded DNA. As a result, telomeres contain **highly repetitive DNA**, with the repeated sequence being 5′-TTAGGG-3′.

(Choice B) Like all DNA in eukaryotes, telomeric DNA *is double-stranded* with each strand complementing the other. Depending on its sequence, eukaryotic RNA (*not* DNA) may self-complement in physiological settings as it is single-stranded.

(Choice D) Restriction enzymes typically cut DNA at sequences known as palindromes, in which the complementary strands are identical when read from 5′ to 3′ ends. The repeating telomere sequence 5′-TTAGGG-3′ is not a palindrome, and therefore it is not susceptible to restriction enzymes.

Educational objective:
Telomeres are highly repetitive stretches of DNA at the ends of a chromosome. They are shortened by each round of DNA replication, and protect the rest of the chromosome from degradation. Because of this, telomeres do not encode any gene products; if they did, the information for those products would be lost upon replication and could harm the cell.

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